



Mathematical
Institute

Holographic Krylov Complexity

From charged AdS probes to top-down quiver geometries

OGOUNCHI Géraud Badélé

MIORPA 2026 Participant

Mentors: Dr Christopher Couzens, Dr Alice Luscher

Mathematical Institute, University of Oxford

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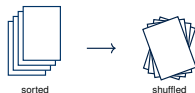


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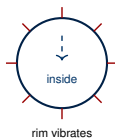


The big picture, in plain words

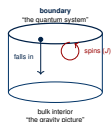
Two very different questions turn out to be the same question.



1. Shuffling a deck of cards



2. A drum



Holography, in one picture

A new deck starts sorted; after many shuffles, it is thoroughly mixed. The **complexity** is a number that tracks how “mixed up” a quantum state has become. Knowing exactly how a drum’s *rim* vibrates lets you reconstruct what happens *inside*; that is **holography**: a quantum system on a boundary is secretly gravity living one dimension deeper, in the **bulk**. This talk: does “falling into the bulk” match “getting mixed up”? Almost, charged particles hide a subtlety, and fixing it is the whole talk.

Quantum states and time evolution: a quick reminder

Quantum mechanics describes the very small. Rather than sitting at a definite place, as in the classical world, a particle has a **probability** of being found somewhere; this is encoded in the **wavefunction**, or **state vector** $|\psi(t)\rangle$, living in a Hilbert space.

- $|\psi(t)\rangle$: the quantum state at time t . Measurement outcomes have probability $|\langle \cdot | \psi(t) \rangle|^2$.
- The **Hamiltonian** H generates time evolution through the Schrödinger equation

$$i \partial_t |\psi(t)\rangle = H |\psi(t)\rangle \quad \implies \quad |\psi(t)\rangle = e^{-iHt} |\psi(0)\rangle.$$

- This evolution is **unitary**: it preserves the norm, $\langle \psi(t) | \psi(t) \rangle = 1$ for all t , so total probability is conserved – the state moves, but nothing is created or lost.
- To describe how $|\psi(t)\rangle$ evolves concretely, we expand it in an orthonormal **basis** of states built from H and $|\psi(0)\rangle$. That basis – the **Krylov basis** – is introduced next.

Krylov complexity: the quantum baseline

Definition

Intuitively, $C(t)$ measures how far $|\psi(t)\rangle$ has spread through the Krylov chain: the effective number of Krylov states needed to describe it, i.e. a measure of **operator/state growth complexity**.

$$|\psi(t)\rangle = \sum_n \phi_n(t) |K_n\rangle, \quad C(t) = \sum_{n=0}^{\infty} n |\phi_n(t)|^2.$$

- **Lanczos algorithm** generates the orthonormal **Krylov basis** $\{|K_n\rangle\}$ by repeated action of H on the initial state, $|K_n\rangle \propto H^n |\psi(0)\rangle$ (Gram–Schmidt orthonormalised); the b_n are the nearest-neighbour Lanczos coefficients.
- Initial condition: $|\psi(0)\rangle = |K_0\rangle$.
- Unitarity implies $C(t) = C(-t)$ is **even**, so

$$C(t) = b_1^2 t^2 + O(t^4), \quad \dot{C}(t) = 2b_1^2 t + O(t^3).$$

Necessary requirement: $\dot{C}(0) = 0$. Every holographic prescription below is tested against this fact.

Holographic proposal and the puzzle

Holography, briefly: duality between a quantum field theory in d dimensions and gravity in $(d + 1)$ dimensions; the QFT “lives on the boundary” of the gravity theory.

heavy boundary excitation $\leftrightarrow$ massive bulk probe,

$$\dot{C}(t) \sim |P_{\text{proper}}(t)|.$$

Minimal charged system – $f(r)$, $g(r)$, $h(r)$ take the specific form of a black hole metric:

$$ds^2 = -f(r) dt^2 + g(r) dr^2 + h(r) d\varphi^2, \quad \underbrace{L = -m\sqrt{f - g\dot{r}^2 - h\dot{\varphi}^2}}_{\text{point-particle Lagrangian}}.$$

- φ is cyclic, so the conserved charge is $J = P_\varphi = \text{const.}$
- Release the particle radially: $\dot{r}(0) = 0 \Rightarrow P_r(0) = 0$, but $P_\varphi(0) = J$ exactly.

The puzzle

$$|P_{\text{naive}}(0)| = \frac{|J|}{\sqrt{h(r_0)}} \neq 0 \quad (J \neq 0) \quad \Rightarrow \quad \dot{C}(0) \neq 0 \quad \text{instead of} \quad \dot{C}(0) = 0.$$

The fix: fix the charge first

J is a conserved quantity, not an independent dynamical direction, once we work at fixed charge. Legendre-transform only $\dot{\varphi}$: $R = L - J\dot{\varphi}$.

Eliminating $\dot{\varphi}(r, \dot{r}, J)$ gives the compact effective Routhian

$$R(r, \dot{r}; J) = -m_{\text{eff}}(r) \sqrt{f(r) - g(r)\dot{r}^2},$$

$$m_{\text{eff}}(r) = \sqrt{m^2 + \frac{J^2}{h(r)}}.$$

New radial momentum:

$$P_r^{(R)} = \frac{m_{\text{eff}}(r) g(r) \dot{r}}{\sqrt{f(r) - g(r)\dot{r}^2}} \quad \Rightarrow \quad P_r^{(R)}(0) = 0 \quad \text{for any } J.$$

Key point

The charge is not discarded. Its effect survives as an effective mass $m_{\text{eff}}(r)$.

Analytical short-time result

Initial data: $r(0) = r_{UV}$, $\dot{r}(0) = 0$, so

$$r(t) = r_{UV} + \frac{1}{2}r_2t^2 + O(t^4), \quad \dot{r}(t) = r_2t + O(t^3).$$

Substituting into $P_r^{(R)} = \partial R / \partial \dot{r}$ gives, to leading order, **Result**

$$P_r^{(R)}(t) = A(J, m, r_{UV}) t + O(t^3)$$

and therefore, via $\dot{C}(t) \sim |P_r^{(R)}(t)|$, **Consequence**

$$\dot{C}(t) \sim t, \quad C(t) \sim t^2$$

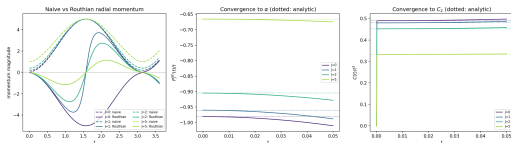
This resolves the puzzle: $P_r^{(R)}(0) = 0$ follows directly from $\dot{r}(0) = 0$, for any J . The naive momentum, by contrast, retains a J -dependent piece even when $\dot{r} = 0$.

Explicit AdS₃ validation

Global AdS₃: $f = 1 + r^2$, $g = 1/f$, $h = r^2$. Solving the Routhian dynamics near the boundary gives a closed-form analytical prediction: **Prediction for $C_2 = b_1^2$**

$$C_2 = \frac{\lambda}{2} \cdot \frac{|J^2 - m^2 r_0^4|}{r_0^2 \sqrt{(m^2 r_0^2 + J^2)(1 + r_0^2)}} \geq 0 \quad \text{for any } (m, J, r_0).$$

- Positivity survives only because we used $P_r^{(R)}$, not P_{naive} ; $J \rightarrow 0$, $r_0 \rightarrow \infty$ gives $C_2 \rightarrow \lambda m/2$, as expected for a neutral heavy operator.
- Numerical check (RK45 integration of the geodesic equation): the analytic slope and C_2 agree with the numerics to ~ 1 part in 10^5 .



From toy model to a genuine 6d SCFT geometry

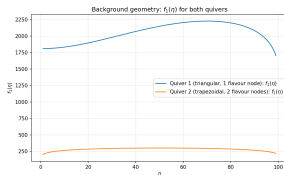
The next test is not another toy model: the massive IIA dual of a 6d $\mathcal{N} = (1, 0)$ SCFT quiver (Fatemiabhari et al. 2026),

$$ds_E^2 = f_1(\eta) ds_{\text{AdS}_7}^2 + f_2(\eta) d\eta^2 + f_3(\eta) d\Omega_2(\theta, \varphi),$$

with the probe moving along (r, η, φ) :

- r : radial AdS₇ coordinate, non-cyclic.
- η : quiver direction, set by the rank function $R(\eta)$, non-cyclic.
- φ : R -symmetry circle, cyclic, with $J = P_\varphi$ conserved.

Structural change: the toy model had one non-cyclic coordinate; this background has two, (r, η) .
 The same puzzle reappears for $J \neq 0$, and the same fixed-charge Routhian resolves it.



Two quivers: the key new validation

Quiver 1 (ranks $N, 2N, \dots, PN$): already studied.

Quiver 2 (constant rank, two flavour nodes): derived and validated independently by the author

$$P_{\text{naive}}(0) = \frac{|J|}{\sqrt{f_3(\eta_0)}} \neq 0 \quad \text{while} \quad P_r^{(R)}(0) = P_\eta^{(R)}(0) = 0 \quad (\text{any } J).$$

Quiver	J	$P_{\text{naive}}(0)$	$P_r^{(R)}(0), P_\eta^{(R)}(0)$
1	0, 1, 3	0, 0.088, 0.263	≈ 0 for every J
2	0, 1, 3	0, 0.231, 0.692	≈ 0 for every J

What is universal vs. what depends on the geometry?

Universal mechanism

- cyclic φ , conserved J
- naive momentum has a nonzero, J -dependent intercept
- fixed-charge Routhian removes the cyclic dynamics
- $P_r^{(R)}(0) = P_\eta^{(R)}(0) = 0$
- $P^{(R)} \sim t$, so $\dot{C} \sim t$, $C \sim t^2$

Geometry-dependent

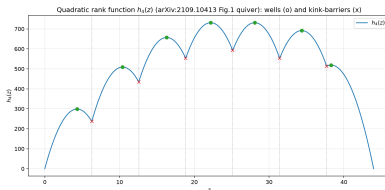
- explicit coefficient A
- effective mass profile $m_{\text{eff}}(\eta)$
- equilibrium position in η
- detailed trajectory
- extent of the clean linear regime

Numerical observation: for Quiver 1 the extracted short-time slope shows a small (2–3%) deviation from the analytic A . This is a finite numerical-window effect: η_0 sits farther from the $J=0$ equilibrium in Quiver 1 than in Quiver 2. It is not a failure of the analytical result, since the strict $t \rightarrow 0$ limit still matches A exactly.

A third background: the black-string-chain quiver

A quiver with a *quadratic* rank function (Couzens et al. 2021), generalizing the piecewise-linear rank functions used above. Fractional D4-branes source $\partial_z^2 h_4 = -\gamma$, giving on each segment $h_4(z) = \alpha + \beta z - \frac{\gamma}{2} z^2$.

- This turns a single potential hump into a multi-well landscape, with local minima at the flavour-brane nodes: the “kinks” seen below.
- The same Routhian construction applies unchanged: $P_r^{(R)}(0) = 0$ for any J , and the exact identity $P_r^{(R)}(t) = Ht$ again holds, to 10^{-12} – 10^{-13} .
- Can the probe be driven from one well into the next? **No**. For every allowed quiver ($\gamma \geq 2$, even) the displacement is 4 to 7 orders of magnitude too small to cross a barrier of order 1.



Conclusion and next step

Main result

Fix the conserved charge before constructing the proper momentum.

1. Krylov theory requires $C(t) \sim t^2$, $\dot{C}(0) = 0$.
2. The naive momentum prescription violates this for $J \neq 0$.
3. The fixed-charge Routhian removes the cyclic motion while retaining its energetic effect through an effective mass.
4. The mechanism survives beyond the toy model: validated in two independent 6d SCFT quivers and in a third, structurally distinct $\text{AdS}_3 \times S^2$ background.

Thank you.

Backup: the naive-momentum intercept, isolated

$P_{\rho}^{\text{naive}}(t)$ and $P_{\rho}^{(R)}(t)$ both grow like $\sim 100 t$, which makes the small nonzero intercept $P_{\text{naive}}(0)$ (only 0.09–0.69, see the table two slides back) invisible if plotted directly. Subtracting the two isolates it: $\Delta P(t) = P_{\rho}^{\text{naive}}(t) - P_{\rho}^{(R)}(t) \rightarrow |J|/\sqrt{f_3(\eta_0)}$ as $t \rightarrow 0$ (dotted lines), confirming the nonzero naive intercept for $J \neq 0$.

The naive-momentum intercept, isolated: $\Delta P(t) \rightarrow |J|/\sqrt{f_3(\eta_0)}$ as $t \rightarrow 0$

